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# The Campanian–Maastrichtian magnetostratigraphy of the Volga region (*vicinity of Volsk town*)

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**Abstract**

Paleomagnetism and rock magnetism have been studied for the first time in Campanian–Maastrichtian carbonate deposits (Sengilei, Karsun, and Radishchev Formations) from the Bolshevik and Kommunar quarries near the town of Volsk (Saratov area, Volga region). Characteristic components of normal and reverse polarity were detected in samples demagnetized in the alternating field, and positive results of the reversal test were obtained. The virtual geomagnetic pole wander paths calculated for the Sengilei Formation and lower Karsun Formation rocks showed statistical similarity to those for stable Europe of the respective age. The magnetostratigraphic zones revealed in the Bolshevik and Kommunar sections were correlated to chrons 33r, 32n2, 32n1, and 31r of the geomagnetic polarity time scale. The magnetic zones of predominantly reverse and normal polarity detected, though discontinuously, in the low magnetic uppermost Karsun Formation and in the Radishchev Formation correspond to the top of chron 31r and joint chrons 31n and 30n, respectively. Rock magnetic data help identifying deposits rich in cosmogenic metal particles and provide stratigraphic constraints on the Radishchev Formation base.
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*Keywords:* paleomagnetism; magnetostratigraphy; Campanian; Maastrichtian; cosmogenic particle; Volga region

## Introduction

Upper Cretaceous deposits in the vicinity of Volsk town (Volga region) have been studied for more than a century but still remain attractive targets for geoscientists. They are perfectly exposed in several cement quarries and contain abundant fossils: belemnite, bivalve, echinoid, ammonite, and brachiopod macrofossils, as well as nannoplankton, plankton, and benthic foraminifera microfossils (Olfer'ev et al., 2009a,b, 2014). Although being exhaustively documented, the sections have had no paleomagnetic and magnetostratigraphic constraints which are indispensable for many practical and theoretical applications: updating local stratigraphy and legends for new geological maps and isochron regional cross correlations, and calibration of zonal scales in different paleobiogeographic provinces.

Integrate biostratigraphic, magnetostratigraphic, lithological, mineralogical, and isotopic studies of Campanian–Maastrichtian reference sections in the southeastern Russian Plate (Guzhikov et al., 2014) were performed as part of project No. 12-05-00196-a of the Russian Foundation for Basic Research (*Stratigraphic Boundaries of the Terminal Cretaceous. The Campanian–Maastrichtian of Eurasia: Global Correlation Problem*), by a large team of geologists from the Saratov and Moscow Universities and the Geological Institute (Moscow). The work aimed at high-resolution Boreal–Tethyan correlation of the Campanian–Maastrichtian boundary in sections of the Russian Plate, by analogy with the Global Boundary Standard-stratotype Section and Point (GSSP) of the Tercis les Bains section in southwestern France (Odin and Lamaurelle, 2001). Paleomagnetc research in the northern Saratov area on the Volga right side was undertaken in 2012–2014 in cement quarries of Bolshevik (52°03′16″ N, 47°25′31″ E) and Kommunar (52°04′25.7″ N, 47°30′10″ E) spaced at 5 km in the northern outskirts of Volsk (Fig. 1). Samples for different types of analysis were collected parallel with geological description and layer-by-layer sampling of macrofossils. About 70 m and 55 m of Campanian–Maastrichtian deposits were documented in the Bolshevik and Kommunar quarries, including oriented large-scale samples from 122 and 74 stratigraphic levels, respectively (Fig. 2).

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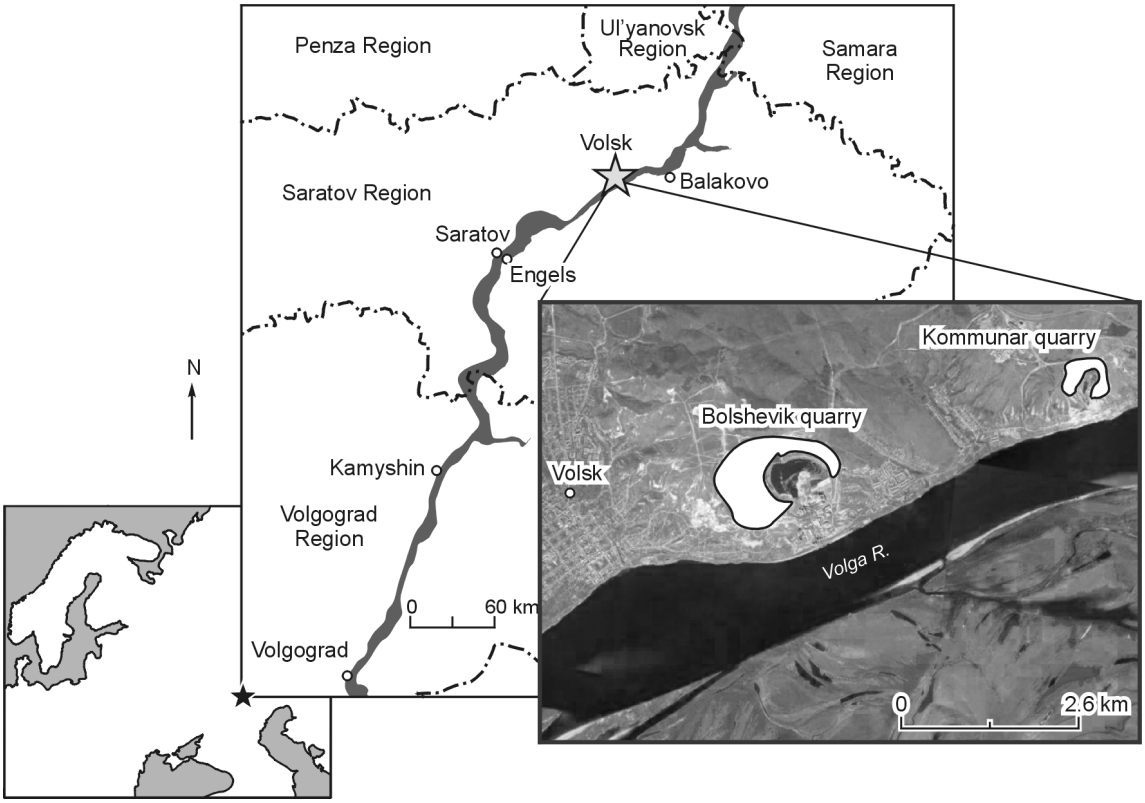



Fig. 1. Location map of sampled sections.

## Stratigraphy

In an earlier publication, Olfer'ev et al. (2009a,b, 2014) provided a detailed historic overview of research on Upper Cretaceous deposits in the vicinity of Volsk and an exhaustive description of the Upper Cretaceous Bolshevik section. Therefore, we confine ourselves to a brief summary of lithology and stratigraphy of the sampled sections.

Upper Cretaceous sediments near Volsk are about 100 m thick and consists of carbonates (chalk-like marl and writing chalk, with lower or higher clay contents) which concordantly lie over Albian black shale (Paramon Formation) and under Paleocene silica clay (Syzran Formation). The Campanian–Maastrichtian deposits belong to the Sengilei, Karsun, and Radishchev Formations (Fig. 2). The Sengilei Formation (7.5 m and 9.0 m thick in the Bolshevik and Kommunar quarries, respectively) has distinct boundaries: its base is delineated by a phosphorite bed and the top of nodular chalk (flaser limestone) is clearly detectable against the overlying marl (Fig. 2). The Radishchev Formation base is marked by a layer of greenish-gray clayey to chalky marl with foraminifera of the Brotzenella praeacuta—Hanzawaia ekblomi zone (Zhamoida, 2004). The carbonate sequence (about 60 m of visible thickness in the Bolshevik section) sandwiched between the Sengilei and Syzran Formations includes tens of such clay layers, and the Karsun–Radishchev Formations boundary is hard to identify visually without microfossil data.

The Sengilei Formation spans the Lower Campanian; the lower Karsun Formation is Upper Campanian and Lower Maastrichtian (greatest part); and the Radishchev Formation is Upper Maastrichtian. The age constraints are due to macofossil (belemnites, bivalves, echinoids, ammonites, and brachiopods) and microfossil (nannoplankton, plankton, and benthic foraminifera) faunas (Guzhikov et al., 2014; Olfer'ev et al., 2009a,b). The Bolshevik and Kommunar quarry sections are almost identical in terms of lithology, paleomagnetism, and magnetism of rocks, but the latter section misses the Radishchev Formation. The deposits generally show planar bedding, though there are landslides in both quarries, especially in the Kommunar one, which probably resulted from active neotectonic uplift of the territory.

## Methods and results

The studies of rock magnetism included measurements of magnetic susceptibility ($K$), natural remanent magnetization (NRM) or remanence ($\mathbf{J}_n$), and magnetic saturation, as well as differential thermomagnetic analysis (dTMA). The following instruments were used: an MFK1-FB kappa bridge for magnetic susceptibility ($K$); a JR-6 spinner magnetometer and a 2G-Enterprises SQUID cryogenic magnetometer for remanence ($\mathbf{J}_n$) (at Institute of Earth's Physics, Moscow); and a TAF-2 fraction thermoanalyzer for dTMA.

The deposits are very low magnetic or often diamagnetic. Magnetic susceptibility ($K$) varies from $-0.54 \times 10^{-5}$ to $1 \times 10^{-5}$ SI units, except for the lower section part (Sengilei and

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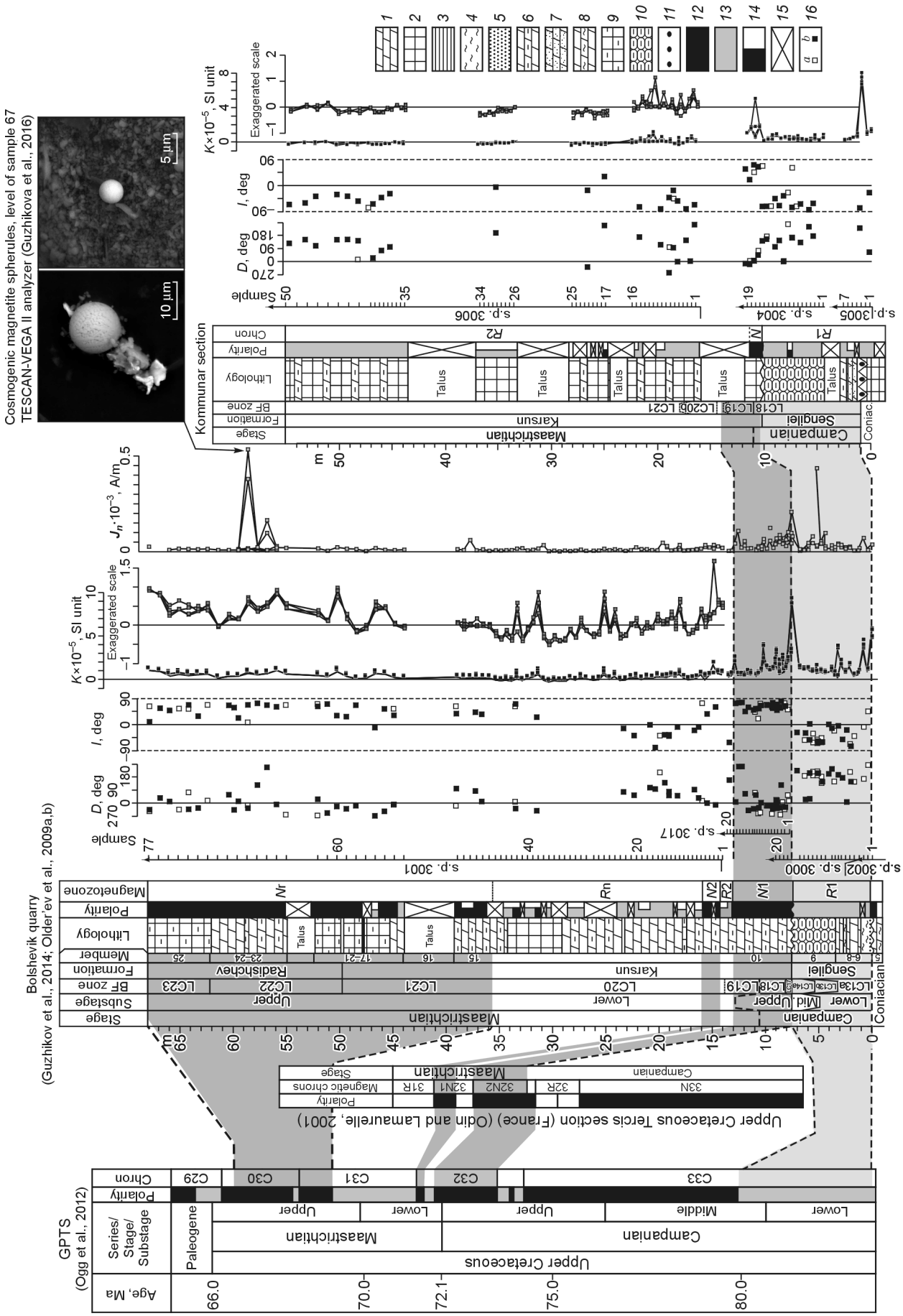



Fig. 2. Campanian–Maastrichtian deposits near Volsk town: magnetostratigraphic correlation with the Maastrichtian GSSP (Odin and Lamaurelle, 2001) and GPTS (Ogg et al., 2012) reference. *1*, marl; *2*, chalk; *3*, clay; *4*, silty material; *5*, sandy material; *6*, clayey marl; *7*, sandy marl; *8*, silty marl; *9*, clayey chalk; *10*, nodular chalk (flaser limestone); *11*, phosphorite; *12*, normal polarity; *13*, reverse polarity; *14*, polarity inferred from low-quality paleomagnetic data; *15*, absence of polarity data; *16*, results of component analysis after measurements on R-6 (*a*) and 2G-Enterprices (SQUID) (*b*) magnetometers; s.p., sampling point.

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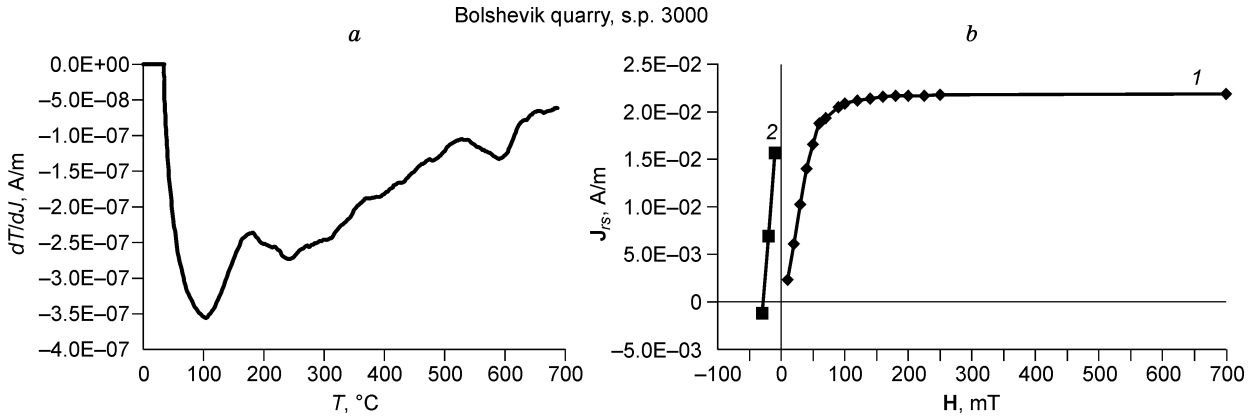



Fig. 3. Magnetic mineralogical analysis. *a*, Differential thermomagnetic curve (sample 16); *b*, typical magnetic saturation/demagnetization curve ($\mathbf{J}_{rs}$ remanent saturation magnetization) (sample 10a): *1*, saturation curve, *2*, demagnetization curve.

lower Karsun Formations) where it reaches $(5–10) \times 10^{-5}$ SI units; NRM ($\mathbf{J}_n$) does not exceed $(0.10–0.27) \times 10^{-3}$A/m (Fig. 2), with rare exceptions (Guzhikova and Bagaeva, 2013). Although being low, the rock magnetic parameters show distinct depth-dependent trends (Fig. 2).

The Sengilei Formation rocks have the highest magnetic susceptibility (to $3.9 \times 10^{-5}$ SI units) and remanence (to $0.43 \times 10^{-3}$A/m). The formation top is marked by anomalous $K$ and $\mathbf{J}_n$ values (to $9.3 \times 10^{-5}$ SI units and $0.21 \times 10^{-3}$A/m, respectively), most likely due to high contents of ferromagnetic particles in the gap zone. Rocks of the lower Karsun Formation are likewise low magnetic but have positive magnetic susceptibility reaching $(4–5) \times 10^{-5}$, which decreases gradually upsection, until the rocks become mainly diamagnetic (to $-0.54 \times 10^{-5}$ SI units) from about the level of sample 3001/7. In the Bolshevik section, $K$ increases from the 3001/54 level and reaches positive values, while the upper section part comprising the upper Karsun Formation and the Radishchev Formation is mostly paramagnetic (Fig. 2).

Unfortunately, most of the dTMA results are of poor information value (possibly, because the signal from the magnetization carrier is beyond the sensitivity of the TAF-2 analyzer), but a few thermomagnetic curves show peaks about the Curie temperature of magnetite, 578 °C (Fig. 3*a*). The magnetic saturation results (Fig. 3*b*) record a magnetically soft phase like fine-grained magnetite in all samples (saturation occurs within 100 mT; residual coercive force varies from 15 to 30 mT (Guzhikova and Bagaeva, 2013)). The hypothesis that fine magnetite is the principal magnetization carrier is consistent with electron microprobe (EMPA) results (analysis performed by V. Tselmovich on a Tescan VEGA II analyzer) that revealed magnetite spherules and iron flakes (Fig. 2), possibly, of cosmogenic origin (Guzhikova et al., 2016).

Paleomagnetic studies followed the standard procedure (Molostovskiy and Khramov, 1997). The remanence $\mathbf{J}_n$ was measured on a JR-6 spinner magnetometer in samples demagnetized by alternating field on an LDA-3 AF demagnetizer. At least two samples were used from each stratigraphic level (four cubic samples $2 \times 2 \times 2$ cm were precut from each oriented large-scale sample). Paleomagnetic directions in replica samples were determined on a 2G-Enterprises SQUID magnetometer at the Institute of Earth's Physics (Moscow). The results of demagnetization and $\mathbf{J}_n$ measurements on the spinner and SQUID magnetometers showed good agreement, while the latter gave Zijderveld diagrams of better quality (Fig. 4). The component analysis of the collected data was performed using the *Remasoft 3.0* software.

In spite of low $\mathbf{J}_n$, the characteristic components of remanent magnetization (**ChRM**) were detected reliably in many samples, mainly in high-magnetic samples of the Sengilei Formation and lower Karsun Formation rocks (Fig. 5*a*). The **ChRM** form regular clusters oriented either in the northern direction on the lower hemisphere or in the southern direction on the upper hemisphere (Fig. 6*a*), which correspond to normal (*N*) or reverse (*R*) polarity of the geomagnetic field. The *N* and *R* components (Fig. 6*a*), as well as all **ChRM** components normalized to normal polarity (Fig. 6*b*), were used to reconstruct virtual geomagnetic pole wandering paths (Fig. 6*c*, *d*).

Note that clustering of paleomagnetic directions is more than twice denser in the samples corresponding to normal polarity (Fig. 6*a*, *d*). Most likely, scattering of *R* vectors results from different degrees of **ChRM** contamination by secondary magnetization from the present geomagnetic field which cannot be completely removed. However, the same effect causes almost no influence on the paleomagnetic statistics for *N* samples because the normal-polarity paleomagnetic field and the present field remagnetization have similar directions.

The Zijderveld diagrams for the low-magnetic samples of the Karsun and Radishchev Formations are mostly of bad quality (Fig. 5*b*). Nevertheless, the $\mathbf{J}_n$ components are detectable, though with high variance, either on the upper (Fig. 7*a*) or lower (Fig. 7*b*) hemispheres, and thus correspond to reverse and normal polarity, respectively. Although the paleomagnetic directions show large variance (possibly, aggravated by large measurement errors for low-magnetic samples, as well as by landsliding in the Kommunar section not taken into account), the polarities of the samples fall into two distinct groups of normal (*N*) or reverse (*R*) zones (Fig. 2). Some samples, for

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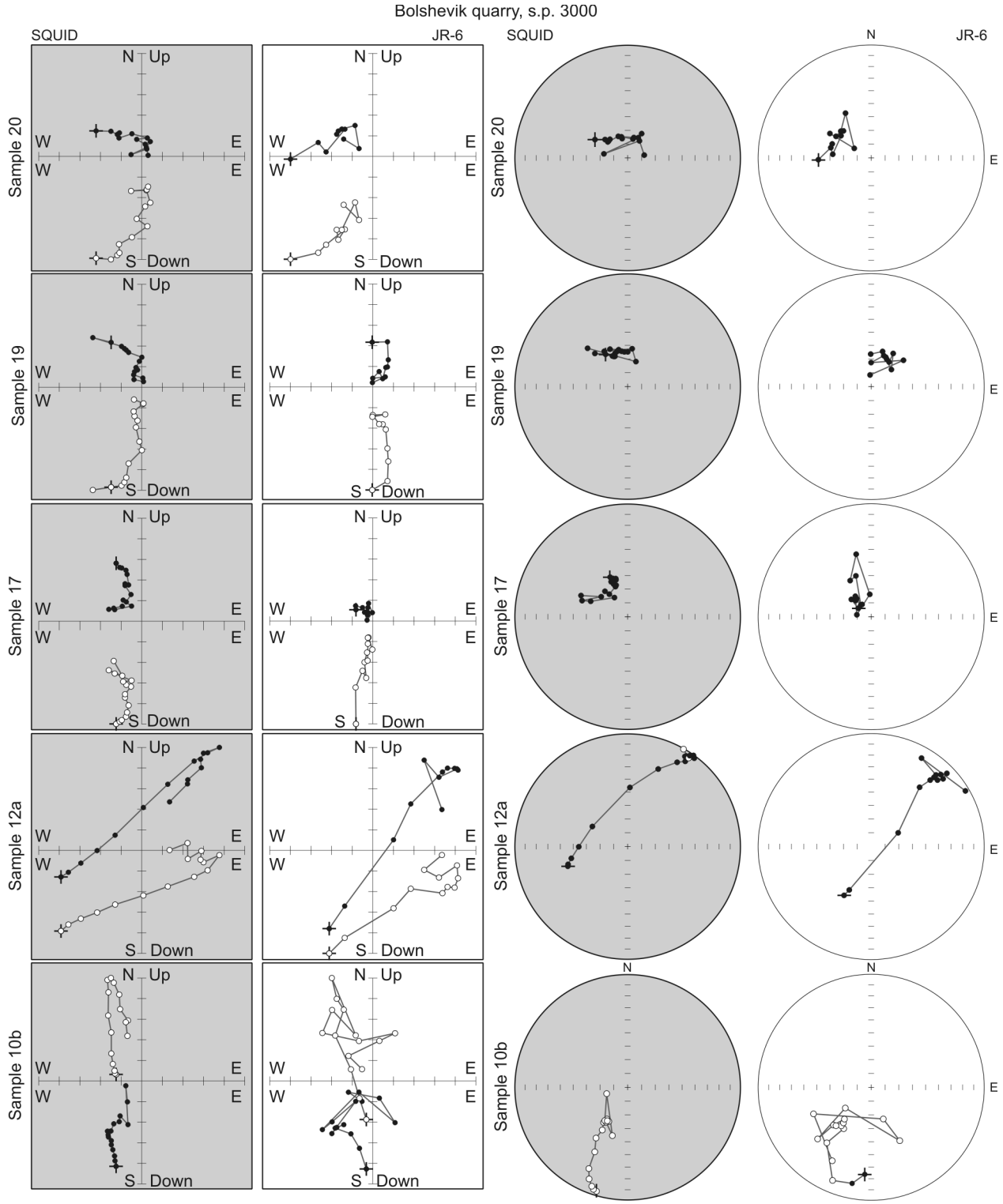



Fig. 4. Results of component analysis with a JR-6 and a 2G-Enterprises SQUID magnetometers (at Saratov State University, Saratov, and Institute of Earth's Physics, Moscow, respectively): compared. Left to right: Zijderveld diagrams (geographic coordinates), stereograms of remanance vector changes during AF demagnetization. Bold and open circles in stereograms are, respectively, lower and upper hemisphere projections. Bold circles are horizontal planes; open circles are vertical planes.

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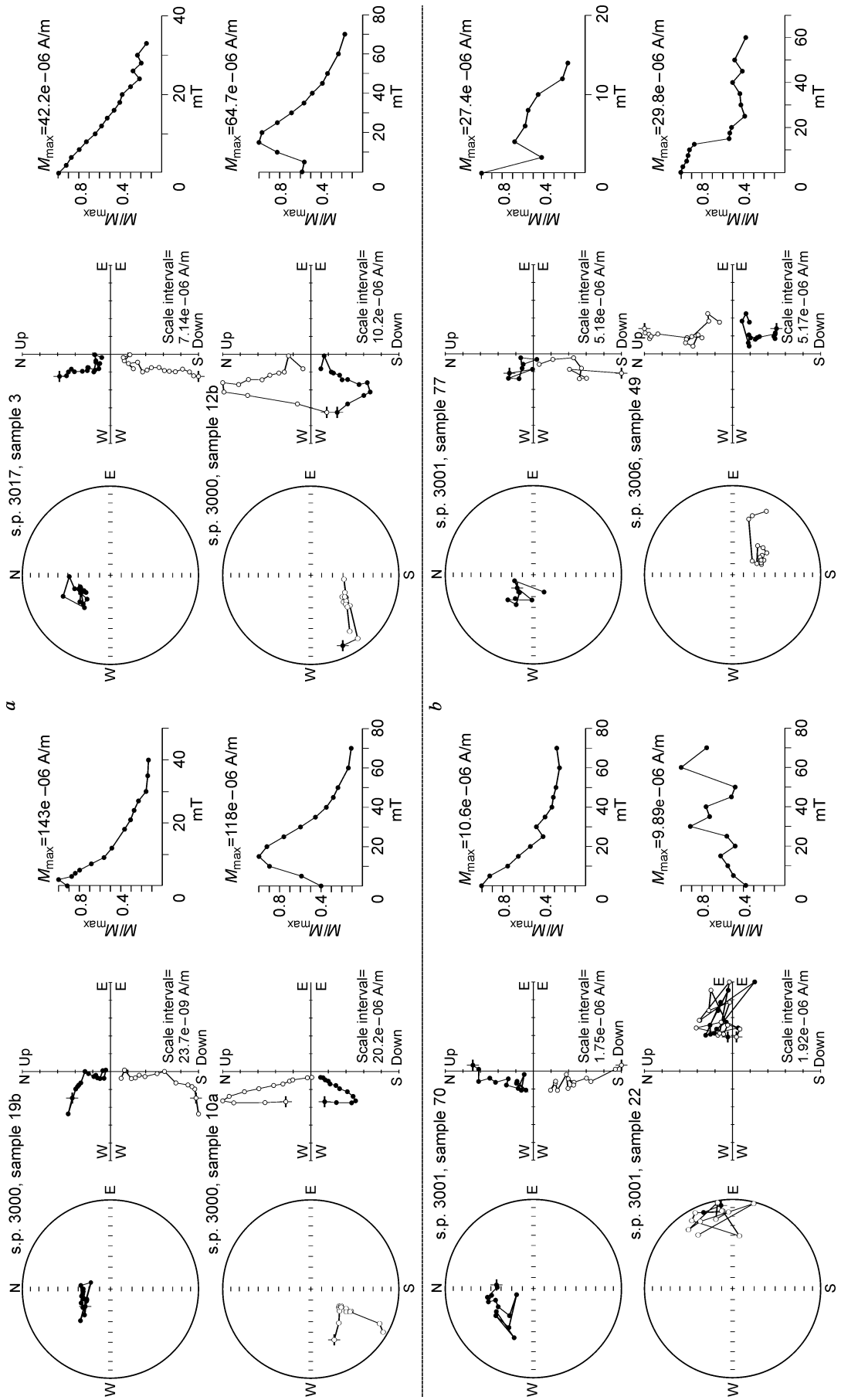
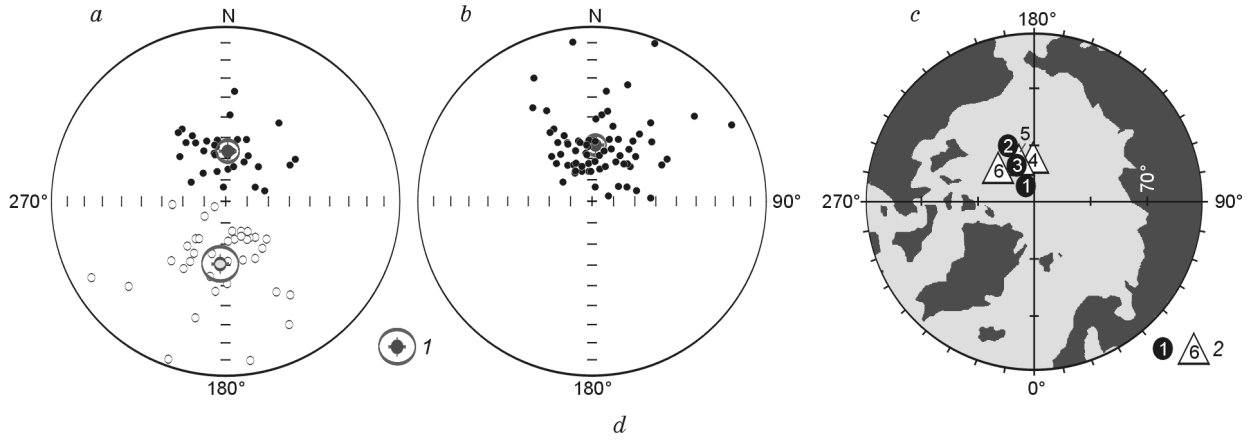



Fig. 5. Results of component analysis. *a*, high-magnetic lower section part (to levels of sample 3001/6 in Bolshevik quarry and sample 3004/19, inclusive, in Kommunar quarry); *b*, upper section part. Left to right: stereograms of remanance vector changes during AF demagnetization; Zijderveld diagrams (geographic coordinates), magnetization variations. Legend same as in Fig. 4.

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*d*

| Parameter | | *n* | $D_{av.}$ | $I_{av.}$ | *k* | $\alpha_{95}$ | Lat | Lon | *dp* | *dm* | $B_{95}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | *N* | 36 | 1.3 | 66.5 | 21.9 | 5.2 | 86.9 | 211.6 | 8.6 | 7.0 | 7.8 |
| 2 | *R* | 36 | 185.4 | –60.5 | 9.35 | 8.3 | 78.8 | 206.1 | 12.6 | 9.6 | 11.0 |
| 3 | *N*+*R* | 72 | 3.5 | 63.6 | 13.06 | 4.8 | 82.8 | 207.4 | 7.6 | 6.0 | 6.8 |
| 4 | 75.4 Ma (averaged over 5 Ma) (Besse and Courtillot, 2002) | | | | | | 81.3 | 188.6 | — | — | — |
| 5 | 77.9 Ma (averaged over 10 Ma) (Besse and Courtillot, 2002) | | | | | | 81.4 | 206.1 | — | — | — |
| 6 | 79.1 Ma (averaged over 5 Ma) (Besse and Courtillot, 2002) | | | | | | 81.0 | 232.5 | — | — | — |

Fig. 6. **ChRM** stereoprojections (geographic coordinates) (*a*, *b*), virtual geomagnetic poles (*c*) and statistical paleomagnetic parameters of **ChRM** distribution (*d*) for the Sengilei and lower Karsun Formations (to levels of sample 3001/6 in Bolshevik quarry and sample 3004/19, inclusive, in Kommunar quarry). *a*, **ChRM** corresponding to normal (*N*) and reverse (*R*) polarities; *b*, **ChRM** normalized to normal polarity. *1*, average paleomagnetic directions and confidence circle; *2*, paleomagnetic pole positions (pole number corresponds to line number in panel *d*); *n* in panel *d* is the number of specimens in a statistical sample; $D_{av.}$, $I_{av.}$ are, respectively, average paleomagnetic declination and inclination (deg); *k* is clustering; $\alpha_{95}$ is confidence circle radius of vectors (deg); Lat, Lon, dp, dm, $B_{95}$ are, respectively, latitude (deg), longitude (deg), semiaxes of error oval and confidence circle radius (deg) for average virtual geomagnetic poles. Results of reversal test (McFadden and McElhinny, 1990): angle of normalized vectors 4.4°, critical angle 10.1°, classification C. Other symbols as in Fig. 4.

which the polarity was unclear anyway, were rejected, but they occur sporadically over the sections and do not affect their general paleomagnetic structure (Fig. 2).

The revealed **ChRM** components are most likely primary, according to common criteria used in the magnetostratigraphy practice (Guzhikov, 2013b):

(1) the presence of normal- and reverse-polarity zones not related to lithology (Fig. 2);

(2) positive reversal test (McFadden and McElhinny, 1990) (Fig. 6*d*);

(3) statistically significant fit of the calculated paleomagnetic polar wander paths to the master APWP (Besse and Courtillot, 2002) (Fig. 6*c*, *d*);

(4) internal consistency of the polarity data obtained from the Volsk sections and their agreement with the general Upper Cretaceous paleomagnetic zoning (Fig. 2), including for West Siberia (Gnibidenko et al., 2012).

The Bolshevik section comprises six paleomagnetic zones (described from bottom to top), each represented by samples from at least four stratigraphic levels (Fig. 2).

*Zone R*1, 7.5 m thick: samples from 19 stratigraphic levels, corresponding to the Sengilei Formation timed reliably as Campanian. The zone obviously correlates with chron 33r of the geomagnetic polarity time scale (GPTS) of Ogg et al. (2012) (Fig. 2).

*Zone N*1, 5.5 m: samples from 25 levels.

Thin *zones R*2 (1.3 m) and *N*2 (1.6 m), samples from three levels for each. Magnetic zones *N*1, *R*2, and *N*2 represent the lower Karsun Formation and correspond to the Campanian–Maastrichtian boundary. They may be equated to GPTS chrons 32n2 and 32n1 (Fig. 2).

*Zone R*n (mostly of reverse polarity, though there are some gaps in the paleomagnetic data), Karsun Formation Zone thickness may be from 19.0 to 28.5 m: the zone top is impossible to rigorously constrain from the available data. **ChRM** were revealed in only 9 samples from 38 (or 47) levels, all of reverse polarity. The polarity for 14 levels more was estimated only tentatively, according to shifts of $\mathbf{J}_n$ projections during demagnetization in the stereograms.

*Zone N*r, mostly normal polarity, upper Karsun and Radishchev Formations. Visible thickness may be from 24 to 34 m, depending on the stratigraphic volume of the underlying zone *R*n. Zone *N*r has fewer gaps in polarity identification than *R*n, while the paleomagnetic quality of samples is slightly higher. The polarity has been identified for 22 (or 28) out of 23 (or 32) stratigraphic levels.

Zones *R*n and *N*r cover the Maastrichtian part of the section and can be easily correlated with GPTS chrons due to a relatively simple paleomagnetic structure of the interval: *R*n and *N*r correspond to 31r and 31n–30n, respectively (Fig. 2).

The polarity record of the Kommunar section is continuous in the Campanian interval (Sengilei Formation and base of

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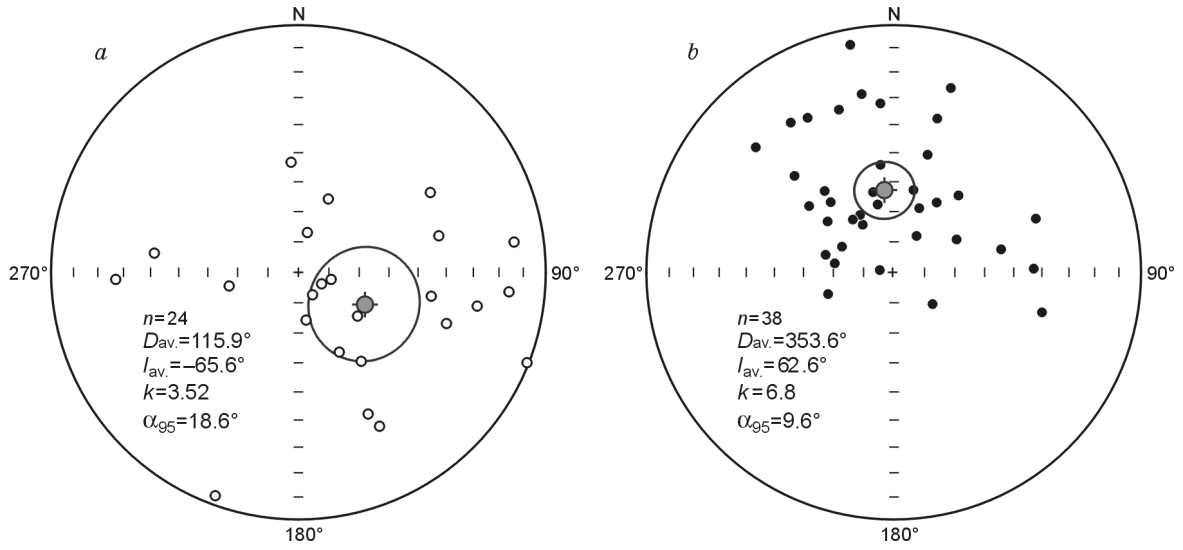



Fig. 7. $\mathbf{J}_n$ stereoprojections (geographic coordinates) corresponding to reverse (*R*) (*a*) and normal (*N*) (*b*) polarities, from samples of low-magnetic section parts (to levels of sample 3001/6 in Bolshevik quarry and sample 3004/19, inclusive, in Kommunar quarry). Legend same as in Figs. 4 and 6.

Karsun Formation) but discontinuous in the Maastrichtian part, because of both uneven exposure and poor polarity identification in low-magnetic samples. All samples of the Sengilei Formation are of reverse polarity while those of the lower Karsun Formation are of normal polarity; reverse polarity predominates over the remaining part of the section (Fig. 2). Thus, there are analogs of the *R*1, *N*1 and *R*n zones in the paleomagnetic structure of the Kommunar section.

The lower *R*-zone coinciding with the Sengilei Formation is 9 m thick and is represented by samples from 20 levels. The *N*-zone at the base of the Karsun Formation is 1.5 m thick and its representative samples are from 4 levels. Paleomagnetic directions for the thickest (at least 39 m) *R*2 zone were determined in 50 samples (41 of reverse polarity).

## Geological implications

1. The analogs of GPTS magnetic chrons 32n2 and 32n1 revealed at the Campanian–Maastrichtian boundary in the Volsk sections, along with fauna evidence, provide rigorous constraints on the Maastrichtian base traceable from the Tercis GSSP (Odin and Lamaurelle, 2001) in SW France to the Volga region. According to biostratigraphic and magnetostratigraphic correlation of the sampled Volsk sections with the GSSP reference, the Tercis Maastrichtian base level falls within zone LC19 (possibly, its lower part) in the Russian Plate (Guzhikov et al., 2014) (Fig. 2).

2. The deposition gap at the Sengilei–Karsun Formations boundary estimated proceeding from paleomagnetic correlation of the Bolshevik and Kommunar quarry sections with GPTS (Ogg et al., 2012), with reference to biostratigraphic data, was at least 6.5 Ma (Fig. 2).

3. Deposition rates can be estimated by dividing the thickness of magnetic zones by the duration of the respective polarity chrons. Thus, marl from the lower Karsun Formation in the Bolshevik quarry spanning zones *R*2 and *N*2 as analogs of GPTS chrons 32*N*1r and 32*N*1n (Fig. 2) was deposited at ~5 and ~7 mm/kyr the slowest, respectively.

4. The base of the Radishchev Formation is marked by benthic foraminifera that trace the Upper Maastrichtian base on the Russian Plate (Zhamoida, 2004). However, these constraints, in a sense, contradict article V.1 of the Stratigraphic Code of Russia (2006) reading that local stratigraphic units are distinguished in local sections according to facies, lithology, and petrography of rocks different from those in the nearby units, both in depth and laterally; local stratigraphic units are identifiable in the field (and boreholes) and mappable.

It appears reasonable to assume that the base of the Radishchev Formation, which has a stratigraphic volume commensurate with a substage, corresponds to the change from diamagnetic to mostly paramagnetic rocks in specific sections. Magnetic susceptibility data can be obtained promptly with a portable KT-10 kappa bridge in the field or in core storage rooms. Variations in rock magnetism reflect the respective structure and texture changes in paramagnetic and ferromagnetic particles and are actually a specific case of lithostratigraphic variations. The use of dramatic changes in magnetic properties, along with lithological, sedimentological, and other criteria, for local stratigraphic division was justified in previous publications (Guzhikov, 2013a; etc.). In this respect, the change from mainly diamagnetic ($K \le 0$) to paramagnetic ($K > 0$) rocks may mark the base of the Radishchev Formation in the Bolshevik section (Fig. 2). Correspondingly, the absence of the Radishchev Formation rocks from the Kommunar section is consistent with predominance of diamagnetic rocks in the upper part, unlike the Bolshevik section where they change to paramagnetic rocks (Fig. 2).

5. Anomalously high magnetization of upper Maastrichtian rocks in the Bolshevik section at the level of sample 3001/67 (Fig. 2) may record the presence of cosmogenic material,

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which has been proven by EMPA data (Guzhikova et al., 2016).

Measurements on a Tescan VEGA II analyzer revealed pure iron, nickel, and magnetite spherules from 3 to 10 μm (Fig. 2) in samples from five stratigraphic levels (several 1 $cm^3$ samples from each level). The number of these particles is 3–6 times greater at the level of sample 3001-67 than at other levels (30–50 against 8–12 particles).

The coincidence of the magnetic anomaly with high contents of cosmogenic material hardly can be fortuitous, and high $\mathbf{J}_n$ values most likely result from especially good ability of cosmogenic particles to align with the geomagnetic field. Their concentration is too small to notably affect $K$ but is sufficient to increase the magnetic moment of sediments (Guzhikova et al., 2016).

The level with high contents of presumably cosmogenic material may be laterally continuous, because similar particles were found earlier by dTMA in Upper Maasatrichtian samples from other sections on the Volga right side: Klyuchi and Teplovka Villages of the Novye Burasy district, Saratov area (Molostovsky et al., 2006).

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